

Comparing First Fit and Best Fit

First Fit

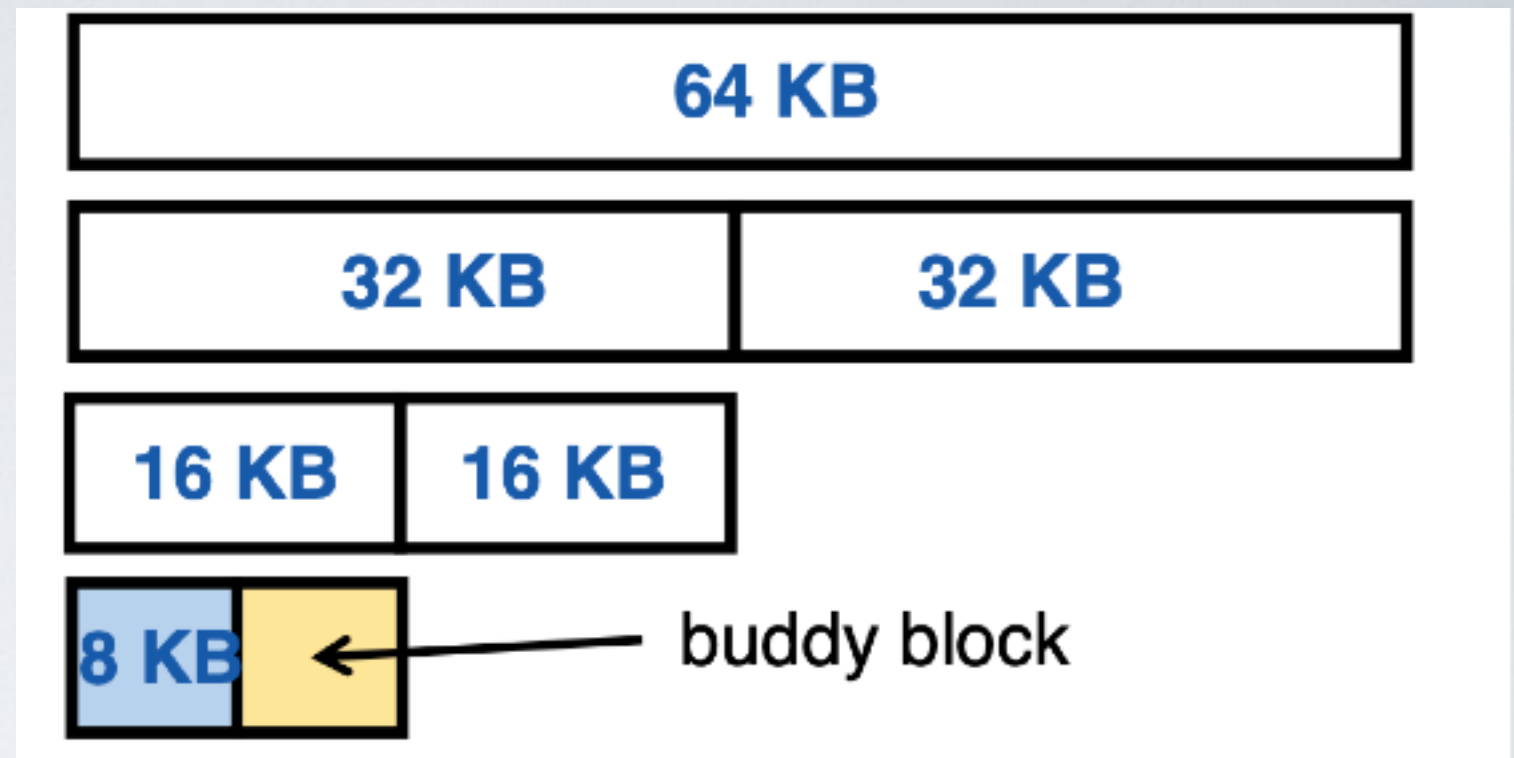
- ✓ Simplest, and often fastest and most efficient
- ⦿ May leave many small fragments near start of memory that must be searched repeatedly

Best Fit

- ✓ In practice, similar storage utilization to first-fit
- ⦿ Left-over fragments tend to be small (unusable)

Buddy Allocation

➡ Allocate blocks in 2^k



Data structure

Maintain n free lists of blocks of size $2^0, 2^1, \dots, 2^n$

Code

- recursively divide larger blocks until reach suitable block
 - insert buddy blocks into free lists
 - upon free, recursively coalesce block with buddy if buddy free
- ➡ the addresses of the buddy pair only differ by one bit